CSCI 1300: Starting Computing – Syllabus

Course Website: https://csci1300.github.io/

Section 300

Instructor: Zachary Kaufman

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Class Time and Location: MTThF 11:10am-12:30pm, ECES 112

Graduate TA: Ishneet Chadha

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Recitation Time and Location: W 11:10am-12:30pm, ECES 112

Office Hours: [TBA]

Section 830

Instructor: Amanda Hernandez Sandate

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Class Time and Location: MTThF 2:50pm-4:10pm, Meets Remotely

Graduate TA: Nolan Brady

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Recitation Time and Location: W 2:50pm-4:10pm, Meets Remotely

Office Hours: [TBA]

Undergraduate Course Assistant: Sashi Wolf

Office Hours: [TBA]

Course Description

Teaches techniques for writing computer programs in higher level programming languages to solve problems of interest in a range of application domains. Appropriate for students with little to no experience in computing or programming.

Course Objectives

This class covers the basics of computer programming using C++. The focus will be on understanding the basic components of computer languages. Upon completion of the course, the student can:

- Design and construct algorithms for problem solving by applying processes of abstraction and program decomposition.
- Implement fundamental programming constructs, such as variables, expressions, conditionals, and iterative control structures, in a higher-level language.
- Evaluate and implement simple I/O ("input/output"), such as user input and file I/O, including the necessity for external input to a program and the role of external data storage.
- Design and explain the role of functions in program construction, including an understanding of parameter passing and return values.
- Describe the properties of data types, including the primitive types of numbers, characters, and booleans, as well as more complex data types, such as arrays, records, and strings.
- Use an Integrated Development Environment to produce code that is free of syntactical, logical, and run-time errors. Understand the process of debugging as part of software development.
- Design and create code using object-oriented design methodology, including an understanding of objects and classes, information encapsulation, and efficient class design.
- Use third-party code to accomplish a programming objective. This includes learning to read code written by another individual and modifying the code or using third-party libraries.
- Develop an understanding that software development is a dynamic, social process, and that learning how to seek out information is a necessary skill for success.

Course Outline

This is a 4-credit-hour course. Students are expected to attend lectures and recitations, stay current with course materials, complete assignments on time, and participate in other course-related activities. Success in the course will require consistent engagement both during scheduled class meetings and outside of class.

Brief list of topics to be covered:

- Computer architecture and environment
- Variables and data types
- Variables and operators
- Strings: indexing, iteration, comparing
- Control structures: if/else statements
- Control structures: switch statements
- Control structures: while loops
- Control structures: for loops, iteration
- Functions
- Arrays
- I/O Streams
- File I/O
- Classes and objects
- Classes, functions, methods
- Inheritance
- Vectors
- Recursion
- References

Grading

Your grade is based on your performance in course activities, including recitations, homework, quizzes, projects, and exams.

Your grade in the course will be based on the following components:

- **Homework (45%).**
    - There will be 7 homework assignments throughout the semester, each weighted equally within the homework category. Each homework assignment will include two parts: Part A and Part B.
        - Part A must be completed during recitation and checked off in person to receive credit. If Part A is not completed and checked off during recitation, it will receive a score of 0. Part A will be worth 5 points.
        - Part B may be completed at home or during recitation if time remains. Part B will be worth 10 points.

        - Homework assignments will be submitted through Gradescope and may include autograded components.
- **Project + Exams (50%).**
    - This category includes the major assessments in the course:
        - *Midterm Exam (100 points):* There will be one midterm exam. This exam will assess your understanding of the course material covered up to that point. The midterm exam may include extra credit opportunities.
        - *Final Project (100 points):* At the end of the semester, you will complete an individual final project that integrates the concepts and skills developed throughout the course. The final project will be graded through an interview-style evaluation and may include extra credit opportunities.
        - *Mid-Project Checkpoints (30 points):* There will be 3 mid-project checkpoint check-ins during the semester, each worth 10 points. These checkpoints are intended to help you stay on track and make steady progress toward the final project.
- **Attendance (5%).**
    - Attendance in lecture is required and contributes to your final grade. Regular attendance is expected so that you can fully participate in course activities and stay current with the material.

Submission Policies

All assignments are due by the date and time specified in the assignment. Homework can be submitted up to two days late for 80% of full credit. After two days, no late work is accepted without prior permission of the instructor.

All activities will be managed through Canvas. The due dates are specified for every assignment.

Homework

Homework will be due every week at 11:59 pm on Tuesdays. Project check-in assignments will be due every week at 11:59pm on Fridays (starting week 5). Coding assignments will often be given immediate feedback using an auto-grade feature, but may also involve a grading interview process outlined below.

Homeworks are divided into two parts, Part A and Part B. Part A must be completed during the recitation section to receive credit for that portion of the assignment. Part B is due the following week on Tuesdays at 11:59pm. Final grade for the assignment will be the completion and correctness of both Part A and Part B.

Homework Late Policy: All homework assignments may be turned in up to 2 days (48 hours) late with a 20% penalty. (Example: If a student would have received a 95% had they turned in their homework on time, a late submission will earn them a 75% instead).

Interview Grading: Following the project submission, each student will sign up for a grading interview about their project. If you are not interviewed for your project, then you get no credit for it. The grading interview will consist of questions about the work you will have submitted the previous week. These questions are designed to test your understanding of the solution code you submitted. They will also serve as an opportunity for you check in about other topics related to the course. It is your responsibility to look on the course website and sign up for an interview slot.

If you have to reschedule a grading interview, you must email the TA at least one day in advance. Documented emergency situations will be evaluated on a case-by-case basis.

Grading Scale

At the end of the course, letter grades will be assigned via the following formula:

100% - 93%	92% - 90%	89% - 87%	86% - 83%	82% - 80%	79% - 77%	76% - 73%	72% - 70%	69% - 60%	Under 60%
A	A-	B+	B	B-	C+	C	C-	D	F

Required Texts & Other Materials

Textbook. No required textbook.

Software. VS Code is the Interactive Development Environment (IDE) we will be using for this course. No other IDE is allowed to be used to ensure everyone is using similar tools, and to make things easier when asking for help. This also means you cannot use IDE’s based on VS Code (such as Cursor).

We will be using the course website https://csci1300.github.io/ to communicate all important dates, lecture recordings, lecture slides, and all other course material.

We will also be using Canvas. Although you only need to do minimal work directly on Canvas, you will need to know how to navigate Canvas to see your course grades.

You will also need to be expected to communicate and collaborate on Discord. You can post your questions in the appropriate channel or thread for each week, assignment, or problem. Your questions may be answered by another student or by a member of the instructional team. Please also answer questions from your classmates when you are able. As a group, we are much more efficient than any one individual.

Other Materials. You must have a back-up system. Lost work, internet was broken, computer froze up, etc. are not valid reasons for missing due dates.

For this course it is highly recommended that you get a Dropbox account, a Google Drive account, invest in a USB memory stick, or GitHub, to save your files. Computer crashes absolutely WILL happen and you are responsible for saving and keeping backup copies of your work. Also, Google Drive and Dropbox keep a versioning system. If you accidentally delete your file, you will be able to recover it.

Collaboration, Plagiarism, and Honor Code

Collaboration is highly recommended and a valuable skill to develop at this point in your life.

Collaboration is defined as working with other people to ask, and answer, questions about how coding is done. This can be accomplished verbally or with pseudo-code or with pictures or with flow charts or with discussions about what syntax means… Collaboration is NOT defined as showing someone your source code, or by cutting-and-pasting code, or by a group project where everyone talks but only one person codes, or by using a paid tutor to show you code, or by using internet sites to copy code.

lthough you must work with other students to fully understand coding, you should never reveal your source code, or copy source code, or leave your browser open, or leave your computer unattended. Stolen code results in ALL involved parties being sent to honor council.

The Computer Science Department at the University of Colorado at Boulder encourages collaboration among students. You are encouraged to work with a partner for some assignments (Final Project), but you are also free to tackle the work alone.

Many forms of collaboration are acceptable and encouraged. In computer science courses, it is usually appropriate to ask others— TAs, LAs, instructor, or other students— for hints and debugging help or to talk generally about problem solving strategies and program structure. In fact, we strongly encourage you to seek such assistance when you need it. Discuss ideas together, but do the coding on your own .

Rule 1: You must not submit or look at solutions or program code that are not your own. Do not search for online solutions. This is not an appropriate way to "check your work," "get a hint," or "see alternative approaches."

Rule 2: You must not share your solution code with other students, nor ask others to share their solutions with you. Do not leave copies of your work on public computers nor post your solution code on a public website.

Rule 3: You must indicate on your submission any assistance you received. It is fine to discuss ideas and strategies, but you should be careful to write your programs on your own.

Be aware: all submissions are subject to automated plagiarism detection. The entire point of the CU Honor Code is that we all benefit from working in an atmosphere of mutual trust. Do not take advantage of that trust. CU employs powerful automated plagiarism detection tools that compare assignment submissions with other submissions from the current and previous quarters. The tools also compare submissions against a wide variety of online solutions. These tools are effective at detecting unusual resemblances in programs, which are then further examined by the course staff. The staff then make the determination as to whether submissions are deemed to be potential infractions of the Honor Code.

To support students in collaboration the Department has created a Collaboration Policy that makes explicit when their collaborative behavior is within the bounds of the Collaboration Policy and when it is actually academic dishonesty, which would be considered a violation of the University's Honor Code. All students of the University of Colorado at Boulder are responsible for knowing and adhering to the University's Honor Code. Violations of this policy may also include cheating, plagiarism, academic dishonesty, fabrication, lying, bribery, and threatening behavior. Collaboration on homework assignments is allowed and encouraged. Students are most successful when they are working with other students to understand new concepts. The ultimate goal is that you fully understand the code you develop.

Plagiarism includes using material from outside sources (e.g. Internet sources, chatGPT, other AI tools, chegg.com, coursehero.com, a tutor) without clear identification and citation. Unless otherwise specified, you may make use of outside resources (internet, other books, other people), but then you must give credit by citing your sources in the comments inside your code.

Citing examples (assuming // indicates beginning of code comment):

// Modified version from https://github.com/Phhere?MOSS-PHP

// Adapted from Program #7.2 in book "Accelerated C++" by Stroustrup

// Worked with Joe Smith from class to come up with algorithm for sorting

//Received suggestions from stackExchange website (see http://...)

//Worked with a tutor on the algorithm for the STORE function

A good rule of thumb: "if it did not come from your brain, then you need to attribute where you got."

Note: you do not need to cite if you are adapting from slides for the course or the required textbook for the course of from the hired staff for the course.

All homework assignments, all quizzes, and all exams will be required to be completed without outside resources. These will be clearly marked as individual assignments. Use of outside resources would violate the collaboration policy. The use of generative AI tools is prohibited on these assignments. This includes, but is not limited to, tools such as ChatGPT, Claude, Gemini, Copilot, Cursor, or any AI-assisted coding. Using Generative AI or any other outside resource on individual assignments is a violation of the course collaboration policy.

Adhering to the Collaboration Policy:

Some examples of violating the collaboration policy include, but are not limited to:

- Sharing a file (source code) with someone else.
- Submitting a file that someone else shared with you.
- Stealing a copy of someone else's work and submitting as your own, even with modification.
- Copying outside resources.
- Using outside resources and not citing your sources.
- Posting on websites like chegg.com. coursehero.com or craigslist.org soliciting a solution to an (or part of an) assignment.
- Soliciting help with commenting your code also constitutes a violation of the collaboration policy.

- Copying solutions from website like chegg.com, coursehero.com, and other websites.
- Using a solution the “tutor” gave you.

Examples of collaborating correctly :

- Sharing pseudo-code.
- Asking another student for a helpful suggestion. Verbally, not written.
- Reviewing another student’s code for bugs/errors.
- Working together on the whiteboard, or paper, to figure out how to approach and solve the problem. In this case you must include that person’s name in your collaboration list at the top of your submission. This includes working with a tutor.

One way to know you are collaborating well is if everyone is developing the code solution individually. This collaboration policy requires that you be able to create the code, or solve the problem on your own before you submit your assignment. You can brainstorm a solution or algorithm with a friend, but the submitted code should not be the same code.

Even if collaboration is stated, it is a violation of the Honor Code to submit effectively identical code with another student or an outside source.

If two or more students submit the same solution, claiming they have been “working with the same tutor”, that constitutes a violation of the Honor Code.

Any discovered incidents of violation of this collaboration policy will be treated as violations of the University’s Academic Integrity Policy and will lead to an automatic academic sanction in the course and a report to both the College of Engineering and Applied Science and the Honor Code Council. **The academic sanction will be an automatic F.**

Note: The instructor reserves the right to change the policy and to apply different academic sanctions if the violation justifies it. Students who are found to be in violation of the Academic Integrity Policy can be subject to non-academic sanctions as well, including but not limited to university probation, suspension, or expulsion.

Collaboration boundaries are hard to define crisply, and may differ from class to class. If you are in any doubt about where they lie for a particular course, it is your responsibility to ask the course instructor.

Students that leave their computers unprotected are also subject to course sanctions mentioned above. When similar code is found, ALL parties are subject to sanctions. This includes the person whose code was “taken”. Protect your source code at all times. Do not forget to sign out of public computers or leave your own machine unattended.

Classroom Behavior

Students and faculty are responsible for maintaining an appropriate learning environment in all instructional settings, whether in person, remote, or online. Failure to adhere to such behavioral standards may be subject to discipline. Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression, veteran status, political affiliation, or political philosophy.

For more information, see the classroom behavior policy, the Student Code of Conduct, and the Office of Institutional Equity and Compliance.

Accommodation for Disabilities, Temporary Medical Conditions, and Medical Isolation

If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely manner so that your needs can be addressed. Disability Services determines accommodations based on documented disabilities in the academic environment. Information on requesting accommodations is located on the Disability Services website. Contact Disability Services at 303-492-8671 or dsinfo@colorado.edu for further assistance. If you have a temporary medical condition, see Temporary Medical Conditions on the Disability Services website.

Netiquette

All students should be aware that their behavior impacts other people, even online. I hope that we will all strive to develop a positive and supportive environment and will be courteous to fellow students and your instructor. Due to the nature of the online environment, there are some things to remember.

1. Always think before you write. In other words, without the use of nonverbals with your message, your message can be misinterpreted. So please think twice before you hit submit.

2. Keep it relevant. There are places to chat and post for fun everyday stuff. Do not stray from the discussion in the assigned questions.
3. Never use all caps. This is the equivalent of yelling in the online world. It is not fun to read. Only use capital letters when appropriate.
4. Make sure that you are using appropriate grammar and structure. In other words, I don't want to see anyone writing "R U" instead of "are you". There are people in the class that may not understand this type of abbreviation, not to mention it does nothing to help expand your writing and vocabulary skills. Emoticons are fine as long as they are appropriate. A smile ☺ is welcome, anything offensive is not.
5. Treat people the same as you would face-to-face. In other words, it is easy to hide behind the computer. In some cases, it empowers people to treat others in ways they would not in person. Remember there is a person behind the name on your screen. Treat all with dignity and respect and you can expect that in return.
6. Respect the time of others. This class is going to require you to work in groups. Learn to respect the time of others in your group and your experience will be much better. Always remember that you are not the only person with a busy schedule, be flexible. Do not procrastinate! You may be one that works best with the pressures of the deadline looming on you, but others may not be that way. The same is true for the reverse. The key to a successful group is organization, communication and a willingness to do what it takes to get it done.

Website: http://www.albion.com/netiquette/corerules.htmlLinks to an external site.

Compiled by Melissa Landin, Instructor, Dept. of Communication, Inver Hills Community College, mlandin@inverhills.edu

Preferred Student Names and Pronouns

CU Boulder recognizes that students' legal information doesn't always align with how they identify. Students may update their preferred names and pronouns via the student portal; those preferred names and pronouns are listed on instructors' class rosters. In the absence of such updates, the name that appears on the class roster is the student's legal name.

Honor Code

All students enrolled in a University of Colorado Boulder course are responsible for knowing and adhering to the Honor Code. Violations of the Honor Code may include but are not limited to: plagiarism (including use of paper writing services or technology [such as essay bots]), cheating, fabrication, lying, bribery, threat, unauthorized access to academic materials, clicker fraud, submitting the same or similar work in more than one course without permission from all course instructors involved, and aiding academic dishonesty.

All incidents of academic misconduct will be reported to Student Conduct & Conflict Resolution: honor@colorado.edu, 303-492-5550. Students found responsible for violating the Honor Code will be assigned resolution outcomes from the Student Conduct & Conflict Resolution as well as be subject to academic sanctions from the faculty member. Visit Honor Code for more information on the academic integrity policy.

Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation

CU Boulder is committed to fostering an inclusive and welcoming learning, working, and living environment. University policy prohibits protected-class discrimination and harassment, sexual misconduct (harassment, exploitation, and assault), intimate partner violence (dating or domestic violence), stalking, and related retaliation by or against members of our community on- and off-campus. These behaviors harm individuals and our community. The Office of Institutional Equity and Compliance (OIEC) addresses these concerns, and individuals who believe they have been subjected to misconduct can contact OIEC at 303-492-2127 or email cureport@colorado.edu. Information about university policies, reporting options, and support resources can be found on the OIEC website.

Please know that faculty and graduate instructors have a responsibility to inform OIEC when they are made aware of incidents related to these policies regardless of when or where something occurred. This is to ensure that individuals impacted receive an outreach from OIEC about their options for addressing a concern and the support resources available. To learn more about reporting and support resources for a variety of issues, visit Don’t Ignore It.

Religious Holidays

Campus policy regarding religious observances requires that faculty make every effort to deal reasonably and fairly with all students who, because of religious obligations, have conflicts with scheduled exams, assignments or required attendance. We will make every effort to accommodate your religious obligations provided that you notify us, email csci1300@colorado.edu, well in advance of the scheduled conflict. Whenever possible, you should notify us at least two weeks in advance of the conflict to request special accommodations.

See the campus policy regarding religious observances for full details.

Mental Health and Wellness

The University of Colorado Boulder is committed to the well-being of all students. If you are struggling with personal stressors, mental health or substance use concerns that are impacting academic or daily life, please contact Counseling and Psychiatric Services (CAPS) located in C4C or call (303) 492-2277, 24/7.

Free and unlimited tele-health is also available through Academic Live Care. The Academic Live Care site also provides information about additional wellness services on campus that are available to students.

Canvas Privacy Policy

You can find a copy of the Canvas Privacy Policy on the Instructure Product Privacy Policy page.

Canvas Accessibility Statement

You can find a copy of the Canvas Accessibility Statement on the Accessibility within Canvas page.
